

ECE 332 Fall 2025 — Exam 1 Answer Key

[CS p.1: SECTION] = page 1 of cheat sheet; [CS p.2: SECTION] = page 2.

Two corrected errors from prior answer keys:**(1) Q1(c):** Exam writes “ $\omega = 2 \text{ MHz}$ ” but MHz is a frequency unit. Must convert: $\omega = 2\pi f = 2\pi \times 2 \times 10^6 = 4\pi \times 10^6 \text{ rad/s}$.**(2) Q6:** Exam gives $f = \frac{2}{\pi} \text{ GHz}$, not $\frac{2}{5} \text{ GHz}$. $\omega = 2\pi f = 2\pi \cdot \frac{2}{\pi} \times 10^9 = 4 \times 10^9 \text{ rad/s}$.**Question 1 — Triangular Loop (20 pts)**Equilateral loop in y - z plane, N turns, side a . $\mathbf{B}(t) = B_0(-2\hat{x} + 3\hat{y}) \sin(\omega t)$.**(a) Magnetic flux through one turn.**

(6 pts)

[CS p.1: SURFACE ELEMENT ds]: loop in y - z plane $\Rightarrow ds = \hat{x} da$.[CS p.1: DOT PRODUCTS]: $\hat{x} \cdot \hat{x} = 1$, $\hat{y} \cdot \hat{x} = 0$.[CS p.1: ROTATING LOOP area table]: equilateral triangle, side a : $A = \frac{\sqrt{3}}{4}a^2$.

$$\begin{aligned}
 \Phi &= \int_S \mathbf{B} \cdot d\mathbf{s} = \int_S B_0(-2\hat{x} + 3\hat{y}) \sin(\omega t) \cdot \hat{x} da \\
 &= A(-2 \underbrace{(\hat{x} \cdot \hat{x})}_1 + 3 \underbrace{(\hat{y} \cdot \hat{x})}_0) B_0 \sin(\omega t) \\
 &= -2AB_0 \sin(\omega t) = -2 \cdot \frac{\sqrt{3}}{4}a^2 \cdot B_0 \sin(\omega t)
 \end{aligned}$$

$$\Phi = -\frac{\sqrt{3}}{2} B_0 a^2 \sin(\omega t) \text{ V}\cdot\text{s}$$

(b) EMF induced in the loop.

(6 pts)

[CS p.1: EMF & FARADAY'S LAW]: $V_{\text{emf}} = -N \frac{d\Phi}{dt}$.[CS p.1: $\partial B/\partial t$ TABLE]: $\frac{d}{dt} \sin(\omega t) = \omega \cos(\omega t)$.

$$V_{\text{emf}} = -N \frac{d}{dt} \left(-\frac{\sqrt{3}}{2} a^2 B_0 \sin(\omega t) \right) = \frac{\sqrt{3}}{2} N a^2 B_0 \cdot \frac{d}{dt} \sin(\omega t)$$

$$V_{\text{emf}}(t) = \frac{\sqrt{3}}{2} a^2 B_0 N \omega \cos(\omega t) \text{ V}$$

(c) Current through 1 k Ω at given values.

(8 pts)

[CS p.1: CIRCUIT & WAVE strip] / [CS p.1: ROTATING LOOP]: $\omega = 2\pi f$.Exam says “ $\omega = 2 \text{ MHz}$ ” $\Rightarrow$ treat as $f = 2 \text{ MHz}$, convert first:

$$\omega = 2\pi f = 2\pi(2 \times 10^6) = 4\pi \times 10^6 \text{ rad/s}$$

Given: $N = 10$, $a = 0.08$ m, $B_0 = 0.02$ T, $\omega = 4\pi \times 10^6$ rad/s, $\omega t = \pi/4$, $R = 1 \text{ k}\Omega = 1000 \Omega$.

[CSp.1: UNIT CIRCLE]: $\cos(\pi/4) = \frac{\sqrt{2}}{2}$.

$$|V_{\text{emf}}| = \frac{\sqrt{3}}{2} (0.08)^2 (10) (0.02) (4\pi \times 10^6) \cos\left(\frac{\pi}{4}\right) = 9.85 \times 10^3 \text{ V}$$

[CSp.1: EMF & FARADAY'S LAW (equivalent circuit)]: $I = V_{\text{emf}}/R$.

$$I = \frac{V}{R} = \frac{9.85 \text{ kV}}{1000 \Omega}$$

$$\boxed{I \approx 9.85 \text{ A}}$$

Question 2 — Moving Wire (12 pts)

$a = 15 \text{ cm} = 0.15 \text{ m}$ along $\hat{y}$, $\mathbf{u} = 12\hat{x} \text{ cm/s} = 0.12\hat{x} \text{ m/s}$, $\mathbf{B} = (0.1\hat{x} - 0.05\hat{z}) \text{ T}$.

[CSp.1: EMF & FARADAY'S LAW (Motional EMF)]: $V_{\text{emf}} = \oint (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}$. $d\mathbf{l} = \hat{y} dy$.

[CSp.1: CROSS PRODUCT]: $\hat{x} \times \hat{z} = -\hat{y}$ (cyclic rule reversed).

$$\mathbf{u} \times \mathbf{B} = 0.12\hat{x} \times (0.1\hat{x} - 0.05\hat{z}) = -\hat{y}(0.12)(-0.05) = 6 \times 10^{-3} \hat{y} \text{ V/m}$$

$$V_{\text{emf}} = 6 \times 10^{-3} \int_0^{0.15} dl = 6 \times 10^{-3} (0.15)$$

$$\boxed{V_{\text{emf}} = 900 \mu\text{V} = 0.9 \text{ mV}}$$

Question 3 — Conduction & Displacement Current (12 pts)

$I_c = 2 \sin(\omega t) \mu\text{A} = 2 \times 10^{-6} \sin(\omega t) \text{ A}$, $\sigma = 10^{-4} \text{ S/m}$, $\epsilon_r = 26\pi$.

(a) Electric field E .

(6 pts)

[CSp.1: DISPLACEMENT & CONDUCTION CURRENT]: $E = V(t)/d$, $I_c = \frac{\sigma A}{d} V(t)$.

$$E = \frac{1}{d} V(t), \quad I_c = \frac{\sigma A}{d} V(t) \Rightarrow V(t) = \frac{I_c d}{\sigma A}$$

$$E = \frac{1}{d} \cdot \frac{I_c d}{\sigma A} = \frac{I_c}{\sigma A} = \frac{2 \times 10^{-6} \sin(\omega t)}{(10^{-4}) A}$$

$$\boxed{E = \frac{0.02}{A} \sin(\omega t) \text{ V/m}}$$

(b) Displacement current I_d at $\omega = 10^9 \text{ rad/s}$, $\omega t = \pi/3$.

(6 pts)

[CSp.1: DISPLACEMENT & CONDUCTION CURRENT]: $I_d = C \frac{dV}{dt}$, $C = \frac{\epsilon_r \epsilon_0 A}{d}$.

From (a): $E = V(t)/d \Rightarrow V(t) = E \cdot d = \frac{d \cdot 0.02}{A} \sin(\omega t)$.

$$I_d = \frac{\epsilon_r \epsilon_0 A}{d} \cdot \frac{dV}{dt} = \frac{\epsilon_r \epsilon_0 A}{d} \cdot \frac{d \cdot 0.02}{A} \cdot \omega \cos(\omega t) = \epsilon_r \epsilon_0 \cdot 0.02 \cdot \omega \cos(\omega t)$$

[CS p.1: UNIT CIRCLE]: $\cos(\pi/3) = \frac{1}{2}$.

$$I_d = (26\pi)(8.85 \times 10^{-12})(0.02)(10^9) \left(\frac{1}{2}\right)$$

$$I_d \approx 7.23 \text{ mA}$$

Question 4 — Charge Relaxation (10 pts)

$$\sigma = 10^{-4} \text{ S/m}, \quad \varepsilon_r = 26\pi.$$

(a) Relaxation time constant.

(5 pts)

[CS p.1: CHARGE RELAXATION]: $\tau_r = \varepsilon_r \varepsilon_0 / \sigma$.

$$\tau_r = \frac{\varepsilon_r \varepsilon_0}{\sigma} = \frac{(26\pi)(8.85 \times 10^{-12})}{10^{-4}}$$

$$\tau_r \approx 7.23 \text{ } \mu\text{s}$$

(b) Time to decay to e^{-3} of initial value.

(5 pts)

[CS p.1: CHARGE RELAXATION]: $\rho_v(t) = \rho_{v0} e^{-t/\tau_r}$.

Set $\rho_{v0} e^{-3} = \rho_{v0} e^{-t/\tau_r}$, then take $\ln$ of both sides:

$$\ln e^{-3} = \frac{-t}{\tau_r} \Rightarrow t = -\tau_r \ln(e^{-3}) = 3\tau_r$$

$$t = 3 \times 7.23 \text{ } \mu\text{s} \approx 21.69 \text{ } \mu\text{s}$$

Question 5 — Lossless EM Wave (30 pts)

$$\mathbf{E}(z, t) = 4 \cos\left((6\pi \times 10^8 \text{ rad/s})t - kz - \frac{\pi}{6}\right) \hat{y} \text{ V/m}. \quad \varepsilon_r = 6, \quad \mu_r = 1.$$

(a) Find ω , k , λ , η .

(10 pts)

[CS p.2: PLANE WAVE PARAMETERS (LOSSLESS)]: read ω from the argument; chain $\omega \rightarrow k \rightarrow \lambda \rightarrow \eta$.

$$\omega = 6\pi \times 10^8 \text{ rad/s}$$

[CS p.2: WAVE PARAMS strip]: $k = \omega \sqrt{\mu_r \mu_0 \varepsilon_r \varepsilon_0}$.

$$k = \omega \sqrt{\mu_r \mu_0 \varepsilon_r \varepsilon_0} = (6\pi \times 10^8) \sqrt{(4\pi \times 10^{-7})(6)(8.85 \times 10^{-12})}$$

$$k \approx 15.40 \text{ rad/m}$$

$$\lambda = \frac{2\pi}{k} = \frac{2\pi}{15.40} \quad \lambda \approx 0.408 \text{ m}$$

[CS p.2: WAVE PARAMS strip]: $\eta = \eta_0 / \sqrt{\varepsilon_r}$.

$$\eta = \frac{377}{\sqrt{6}} = \frac{377}{2.449} \quad \eta \approx 153.91 \text{ } \Omega \quad (\text{lossless})$$

(b) Find $\tilde{\mathbf{E}}(z)$.

(5 pts)

[CS p.1: PHASOR METHOD]: $\mathbf{E}(z, t) = \text{Re}\{\tilde{\mathbf{E}}(z) e^{j\omega t}\}$. Strip $e^{j\omega t}$, keep the spatial/phase part.Let $\phi = \pi/6$. From $\mathbf{E} = 4 \cos(\omega t - kz - \phi) \hat{y}$:

$$\tilde{\mathbf{E}}(z) = 4 e^{j(-kz - \phi)} \hat{y} = 4 e^{-j(kz + \pi/6)} \hat{y}$$

$$\boxed{\tilde{\mathbf{E}}(z) = 4 e^{-j(15.40 z + \pi/6)} \hat{y} \text{ V/m}}$$

(c) Find $\tilde{\mathbf{H}}(z)$.

(5 pts)

[CS p.2: DIRECTION OF PROPAGATION]: $-kz$ in argument $\Rightarrow \hat{k} = +\hat{z}$.[CS p.2: LOSSLESS vs. LOSSY — E & H]: $\tilde{\mathbf{H}} = \frac{1}{\eta} (\hat{k} \times \tilde{\mathbf{E}})$.

$$\tilde{\mathbf{H}} = \frac{1}{\eta} (\hat{z} \times \tilde{\mathbf{E}}) = \frac{1}{\eta} (\hat{z} \times 4 e^{-j(15.40 z + \pi/6)} \hat{y})$$

[CS p.2: $\hat{k} \times \hat{E}$ table]: $\hat{z} \times \hat{y} = -\hat{x}$.

$$= -\frac{4}{\eta} e^{-j(15.40 z + \pi/6)} \hat{x} = -\frac{4}{153.91} e^{-j(15.40 z + \pi/6)} \hat{x}$$

$$\boxed{\tilde{\mathbf{H}}(z) \approx -25.99 \times 10^{-3} e^{-j(15.40 z + \pi/6)} \hat{x} \text{ A/m}}$$

(d) Find $\mathbf{H}(z, t)$.

(5 pts)

[CS p.1: PHASOR METHOD]: $\mathbf{H}(z, t) = \text{Re}\{\tilde{\mathbf{H}} e^{j\omega t}\}$.

$$\begin{aligned} \mathbf{H}(z, t) &= \text{Re}\left\{-25.99 \times 10^{-3} e^{-j(15.40 z + \pi/6)} \hat{x} \cdot e^{j\omega t}\right\} \\ &= \text{Re}\left\{-25.99 \times 10^{-3} e^{j(\omega t - 15.40 z - \pi/6)} \hat{x}\right\} \end{aligned}$$

$$\boxed{\mathbf{H}(z, t) = -25.99 \times 10^{-3} \cos\left(6\pi \times 10^8 t - 15.40 z - \frac{\pi}{6}\right) \hat{x} \text{ A/m}}$$

(e) Average power density.

(5 pts)

Lossless medium. [CS p.2: AVERAGE POWER DENSITY (POYNTING) lossless shortcut]: $\mathbf{S}_{\text{av}} = \frac{|E_0|^2}{2\eta} \hat{k}$.

$$\mathbf{S}_{\text{av}} = \frac{|E_0|^2}{2\eta} \hat{z} = \frac{4^2}{2(153.91)} \hat{z}$$

$$\boxed{\mathbf{S}_{\text{av}} \approx 51.98 \times 10^{-3} \hat{z} \text{ W/m}^2}$$

Question 6 — EM Wave in Distilled Water (16 pts) $f = \frac{2}{\pi} \text{ GHz}, \quad \sigma = 10^{-4} \text{ S/m}, \quad \varepsilon_r = 26\pi, \quad \mu_r = 1.$

[CS p.2: WAVE PARAMS strip]: $\omega = 2\pi f = 2\pi \cdot \frac{2}{\pi} \times 10^9 = 4 \times 10^9 \text{ rad/s}$

(a) Find $\varepsilon''/\varepsilon'$.

(6 pts)

[CS p.2: LOSSY MEDIA — CLASSIFICATION]: $\frac{\varepsilon''}{\varepsilon'} = \frac{\sigma}{\omega \varepsilon_r \varepsilon_0}$.

$$\frac{\varepsilon''}{\varepsilon'} = \frac{\sigma}{\omega \varepsilon_r \varepsilon_0} = \frac{10^{-4}}{(4 \times 10^9)(26\pi)(8.85 \times 10^{-12})}$$

$$\frac{\varepsilon''}{\varepsilon'} \approx 34.58 \times 10^{-6}$$

(b) What type of medium?

(5 pts)

[CS p.2: LOSSY MEDIA — CLASSIFICATION table]: $\varepsilon''/\varepsilon' < 0.01 \Rightarrow$ low-loss dielectric.

Since $34.58 \times 10^{-6} \ll 0.01$, at this frequency distilled water is a **low-loss dielectric**.

(c) Skin depth.

(5 pts)

[CS p.2: LOW-LOSS MEDIUM / SKIN DEPTH]: $\delta_s = \frac{2}{\sigma \eta}$.

$$\eta = \frac{\eta_0}{\sqrt{\varepsilon_r}} = \frac{377}{\sqrt{26\pi}} = \frac{377}{9.038} \approx 41.71 \, \Omega$$

$$\delta_s = \frac{2}{\sigma \eta} = \frac{2}{(10^{-4})(41.71)}$$

$$\delta_s \approx 479.5 \text{ m}$$

Score (if perfect): Q1: 20/20 Q2: 12/12 Q3: 12/12 Q4: 10/10 Q5: 30/30 Q6: 16/16 = **100/100**